MAKE THINGS HAPPY !

# 2x AI-ADS1115 + 2x SPDT - IoTextra Combo

## Combined Relay & Analog Input Module



The **2x AI-ADS1115 + 2x SPDT - IoTextra Combo** module (hereinafter **IoTextra Combo**) is a versatile solution that combines the functionality of two distinct systems: high-precision analog data acquisition and power switching via mechanical relays.
The module allows measuring two differential analog signals, both voltage and current. Analog input channels are based on the 16-bit ADS1115 delta-sigma ADC from Texas Instruments. The module eliminates the need for external protection and signal conditioning circuits by integrating professional-grade filtering and stabilization directly into the signal path.
Input power: +5VDC. It includes reverse polarity protection.

This Datasheet covers version 3-04. Version 3-04 includes minor revisions over v3-03; hardware QR code references v3-03. Up-to-date technical information about the module is available on GitHub.

### TECHNICAL ANALOG CIRCUIT DESIGN:

The module uses high-quality operational amplifiers and low-tolerance passive components to minimize noise and ensure measurement accuracy corresponding to the ADC resolution:

- **Reference Foundation (Common for Lite & STD)**: Both versions are equipped with the REF3425 precision voltage reference and TLV333 zero-drift operational amplifiers.
- **Dual Active Filtering**: The signal is filtered at both the Op-Amp input and its output before reaching the ADC.
- **Differential Design**: A fully differential signal path ensures maximum common-mode noise rejection.
- **Clean Power**: The analog section is powered by a dedicated TPS62173 (+5V_AN) DC-DC converter, isolating measurements from digital noise.

### CHOOSE YOUR PRECISION LEVEL:

The difference between versions lies solely in the tolerance and stability of the passive components in the analog section.

| Feature | Lite Precision Level | STD (Standard) Precision Level |
|---|---|---|
| **Resistors (Analog Path)** | Industrial (1%) | Precision (0.1% Low-T.C. ) |
| **Filter Capacitors** | Industrial (Low-ESR) | Precision (High Stability) |
| **Voltage Reference** | REF3425 (VREF) | REF3425 (VREF) |
| **Operational Amplifier** | TLV333 (Zero-Drift) | TLV333 (Zero-Drift) |

### SOFTWARE POWER:

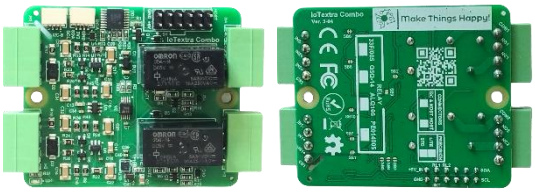
The **IoTextra Combo** module offers native support for popular IoT platforms:

Node-RED TASMOTA Blynk IoTflow

This hardware compatibility unlocks the full potential of the IoTflow platform. It enables you to:

- Create custom **Node-RED** nodes effortlessly.
- Integrate via **MQTT** with third-party global platforms like **Blynk**.
- Scale your project from a single node to a complex industrial network.

Relay channels have galvanic isolation with a dielectric strength 4000Vrms (for one minute between the coil and the contacts) and 750Vrms (for one minute between the opened contacts). Surge resistance between coil and contacts 8000V.
Voltage measurement ranges: 0-0.5V, 0-5V, 0-10V, ±0.5V, ±5V, ±10V.
Current measurement ranges: 0-20mA, 4-20mA, ±20mA, 0-40mA.
Measurement speed: up to 860 samples per second for the **ADS1115**.



## APPLICATIONS:

🏭 **Industrial Telemetry & PLC Integration**
Collects data from 4-20mA and 0-10V sensors in high-EMI environments, while the onboard relays switch small industrial loads directly — a compact, space-efficient alternative to separate analog and relay modules.
*Recommended software: Node-RED, IoTflow*

🌱 **Smart Agriculture**
Monitors soil moisture, pH, and conductivity (EC) sensors while simultaneously controlling irrigation pumps via the onboard relays — sensing and actuation on a single board.
*Recommended software: Node-RED, IoTflow*

🤖 **Robotics & Lab Automation**
Combines signal measurement and load switching in a single test bench — long-term reading stability suits calibration rigs and automated test equipment.
*Recommended software: Node-RED, IoTflow*

🏠 **Smart Home**
Reads sensor feedback (temperature, light level, moisture) on the analog channels while the onboard relays switch lighting, pumps, or small actuators — sensing and control combined on one board.
*Recommended software: Home Assistant*

## FEATURES:

- Compatibility with major known microcontrollers
- Module power supply is 5VDC
- Protection against reverse power supply polarity is included
- Dual-Functionality: 2 Power Relays and 2 Differential Analog Inputs
- 2 **SPDT (1 Form C)** type relays
- Switching Voltage: 250VAC/30VDC
- Contact rating (resistive):

| Relay | NO | NC |
|---|---|---|
| Panasonic ALQ105 | 10A 125VAC<br>5A 250VAC<br>5A 30VDC | 3A 125VAC<br>2A 250VAC<br>1A 30VDC |
| Omron G5Q-14 | 10A 125VAC<br>5A 250VAC<br>5A 30VDC | 3A 125VAC<br>3A 250VAC<br>3A 30VDC |
| HF33F/005-ZS3 | 10A 125VAC<br>5A 250VAC<br>5A 30VDC | 3A 250VAC<br>3A 30VDC |
| TE PE014005 | 5A 240VAC<br>5A 30VDC | |

- Rated operating current is less than 80mA
- Coil resistance - 63Ω
- The dielectric strength of the insulation:
    - 4000Vrms for one minute between the coil and the contacts
    - 750Vrms for one minute between the opened contacts
- Surge resistance between coil and contacts is 8000V
- When working with the module:
    - logical “1” (2.4-5V) – this turns off the relay (NC)
    - logical “0” (0-0.9V) – relay activation (NO)

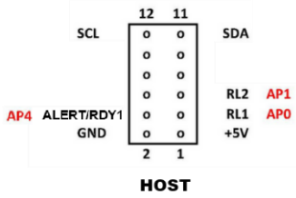
- The module features LED indicators for the relay status: LED lights up when the relay is on
- The relay is controlled using **AP0-AP1** signals (**RL1-RL2**) via the **HOST** connector and/or over the **I²C** serial bus via an I/O expander
- The I/O expander used in the module is a **TCA9534** or a compatible one; its INT output and the ADS1115 ALERT/RDY output share the INT line (pin 4) of the HOST connector
- The expander's **I²C** address (**A2-A0**) is set using jumpers **SB5, SB4** and **SB3** on the **bottom-side** of the module:

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| TCA9534 | 0 | 1 | 0 | 0 | A2 | A1 | A0 | x |
| TCA9534A | 0 | 1 | 1 | 1 | A2 | A1 | A0 | x |

- The default address on the **I²C** bus is 0100111x (for **TCA9534**) or 0111111x (for **TCA9534A**)
- ADC resolution - 16 bits (15 bits effective for unipolar ranges)
- ADC type - ΔΣ (delta-sigma)
- 2 differential analog input channels
- Voltage measurement ranges:
  0-0.5V, 0-5V, 0-10V,
  ±0.5V, ±5V, ±10V
- Current measurement ranges:
  0-20mA, 4-20mA, ±20mA, 0-40mA
- ADC measurement speed up to 860 samples per second (shared by both channels, so up to approx. 430 samples per second per channel when both channels are scanned)
- The **I²C** address of the ADC is changeable: the ADC for channels 1 and 2 has an address of 1001001x (which can be changed to 1001000x using **SB2** on the reverse side of the module)
- Programmable Gain Amplifier (PGA)
- **I²C** interface: 100kHz, 400kHz, 3.4MHz (if supported by the host)
- Connection to the module via the **I²C** bus is done through **Qwiic©** connectors or through pins 11 (**SDA**) and 12 (**SCL**) of the **HOST** connector
- Transient suppression and electrostatic discharge protection of signals on **Qwiic©** connectors is provided by a TVS diode assembly (ESD protection)
- Module size: 47x56 mm. The module has mounting holes that allow it to be installed on a base module or a Raspberry Pi.

## MODULE USAGE AND HOST CONNECTOR

A 12-pin **HOST** connector is always installed on the **IoTextra Combo** module. The connector pins are shown in the following figure




Maximize your project's potential with the Make Things Happy! Ecosystem. The module can be used in the following ways:

1) ***Standalone Use***: Access signals via **HOST** or **Qwiic®** connectors
2) ***Smart Use***. In this mode, an **IoTsmart** module with a microcontroller is plugged into the **HOST** connector. Connection with the **IoTextra Combo** module is through the **HOST** connector. Power is drawn from the **IoTsmart** module. The figure below shows the **IoTextra Combo** module with an **IoTsmart** module:



**IoTextra Combo** with **IoTsmart RP2350** module



**IoTextra Combo** with **IoTsmart ESP32-S3** module

3) ***Mezzanine Use***. In this mode, the **IoTextra Combo** module is installed in a base module, and connection with it is through the **HOST** connector.

## VOLTAGE AND CURRENT MEASUREMENT CALCULATION

The measurement circuit for each channel consists of an operational amplifier, the Programmable Gain Amplifier (PGA) built into the **ADS1115**, and the 16-bit ADC (delta-sigma type) itself. The gain of the operational amplifier depends on the resistance **R**, which can be 24.95kΩ or 49.9kΩ (determined by a pair of jumpers for each channel), and is 24.95kΩ by default. 49.9kΩ improves accuracy only for the 0-0.5V and ±0.5V ranges; for 0-10V, ±10V and 0-40mA it is not allowed (the op-amp output would exceed its usable range).

The output voltage of the operational amplifier, depending on the measured signal range and considering the **+2.5V** reference voltage, will be as follows (the usable output range of the operational amplifier is approx. 0.05…4.95V; with R=49.9kΩ, the ±5V and ±20mA ranges operate close to its lower limit). The ADC measures the differential voltage between the operational amplifier output and the 2.5V reference, i.e. (R/105kΩ)×Vin, therefore the recommended PGA is selected for this differential voltage. In the Recommended PGA column, the left value applies to R=24.95kΩ and the right value to R=49.9kΩ (for the 0.5V ranges with R=49.9kΩ, the signal uses about 93% of the ±0.256V scale; use PGA 8 if over-range detection is required):

| Range | Input Signal | | Op-Amp Output Signal | | Recommended PGA | |
|---|---|---|---|---|---|---|
| | $I_{in}$ | $V_{in}$ | V for R=24.95kΩ | V for R=49.9kΩ | | |
| 0-0.5V | | 0 | 2.5 | 2.5 | 16 | 16 |
| | | 0.5 | 2.618809524 | 2.737619048 | | |
| ±0.5V | | -0.5 | 2.381190476 | 2.262380952 | 16 | 16 |
| | | 0.5 | 2.618809524 | 2.737619048 | | |
| 0-5V | | 0 | 2.5 | 2.5 | 2 | 1 |
| | | 5 | 3.688095238 | 4.876190476 | | |
| ±5V | | -5 | 1.311904762 | 0.123809524 | 2 | 1 |
| | | 5 | 3.688095238 | 4.876190476 | | |
| 0-10V | | 0 | 2.5 | n/a | 1 | n/a |
| | | 10 | 4.876190476 | n/a | | |
| ±10V | | -10 | 0.123809524 | n/a | 1 | n/a |
| | | 10 | 4.876190476 | n/a | | |
| 0-20mA | 0 | 0 | 2.5 | 2.5 | 2 | 1 |
| | 0.02 | 4.98 | 3.683342857 | 4.866685714 | | |
| ±20mA | -0.02 | -4.98 | 1.316657143 | 0.133314286 | 2 | 1 |
| | 0.02 | 4.98 | 3.683342857 | 4.866685714 | | |
| 4-20mA | 0.004 | 0.996 | 2.736668571 | 2.973337143 | 2 | 1 |
| | 0.02 | 4.98 | 3.683342857 | 4.866685714 | | |
| 0-40mA | 0 | 0 | 2.5 | n/a | 1 | n/a |
| | 0.04 | 9.96 | 4.866685714 | n/a | | |

The Full-Scale Range (FS) depends on the gain coefficient (PGA):

| PGA SETTING | FS (V) |
|---|---|
| 2/3 | ±6.144V(1) |
| 1 | ±4.096V(1) |
| 2 | ±2.048V |
| 4 | ±1.024V |
| 8 | ±0.512V |
| 16 | ±0.256V |

(1) This parameter expresses the full-scale range of the ADC scaling. In no event should more than VDD + 0.3V be applied to this device.

To determine the **measured voltage**, the following equation should be used:

$$\text{ADCNumber} = 2^{15} \times \frac{1}{FSR} \times \frac{R}{105k\Omega} \times V_{in}$$

where:

**R** - is the resistance, which can take the value 24.95kΩ or 49.9kΩ depending on the jumpers
**FSR** - is the positive full-scale range, determined by the range setting in the PGA[2:0] configuration register
$\mathbf{V_{in}}$ - is the input voltage

To determine the **measured current**, the following equation should be used:

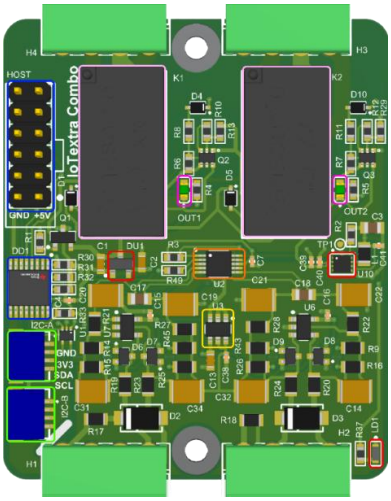
$$\text{ADCNumber} = 2^{15} \times \frac{1}{FSR} \times \frac{R}{105k\Omega} \times 249\Omega \times I_{in}$$

where:

**R** - is the resistance, which can take the value 24.95kΩ or 49.9kΩ depending on the jumpers
**FSR** - is the positive full-scale range, determined by the range setting in the PGA[2:0] configuration register
$\mathbf{I_{in}}$ - is the input current (effective shunt ≈248.7Ω: 249Ω loaded by the 210kΩ amplifier input)

## COMPONENT LAYOUT

The figure below shows the component placement on the **top-side** of the **IoTextra Combo** module:



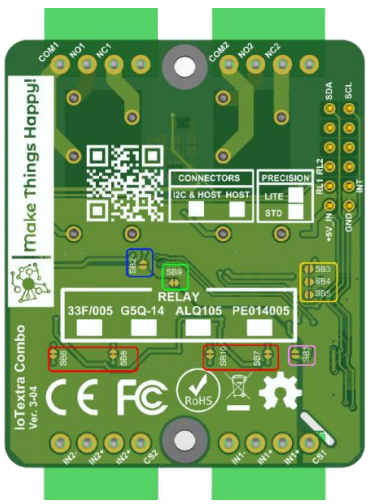

The main **HOST** connector and the [TCA9534](#) I/O expander are highlighted in blue.
The **Qwiic©** connectors are highlighted in salad color.
Relays are highlighted in pink.
Relay status indicators are highlighted in purple.
Highlighted in red are:

- The high-precision DC-DC converter **TPS62173**, used to obtain the analog +5V power
- The low-noise LDO used to obtain +3.3V
- The analog power indicator

The **ADS1115** ADC is highlighted in orange.
The high-precision 2.5V reference voltage source (**REF3425** type) is highlighted in yellow.

## JUMPERS

The figure below shows the component placement on the **bottom-side** of the **IoTextra Combo** module:



The following jumpers are located on the **bottom-side** of the module:

- **SB1** (highlighted in pink) - for connecting +3V3 within the module and the **Qwiic©** connectors. By default, the jumper is open, meaning there is no +3V3 power connection.
- **SB2** (highlighted in light blue) - for selecting the **I²C** interface address of the ADC for analog input channels 1 and 2 (this ADC is **U2** in the schematic). By default, this jumper is open. This means the ADC address is 1001001x. If this jumper is closed, the address of this ADC becomes 1001000x.
- The **SB3**, **SB4** and **SB5** (they are highlighted in yellow) determines the **I²C** address of the I/O expander.

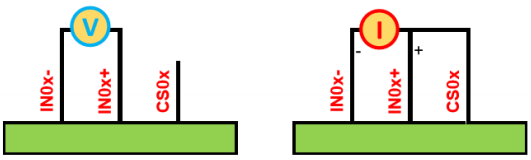
| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| TCA9534 | 0 | 1 | 0 | 0 | A2 | A1 | A0 | x |
| TCA9534A | 0 | 1 | 1 | 1 | A2 | A1 | A0 | x |

  The default address on the **I²C** bus is 0100111x (for **TCA9534**) or 0111111x (for **TCA9534A**)
- For selecting resistances, the jumpers are: **SB7** and **SB10** (channel 1), **SB6** and **SB8** (channel 2). These jumpers are highlighted in red. For each channel, both jumpers in the pair are closed by default, meaning **R** equals 24.95kΩ (see above). If both jumpers in the pair are opened, **R** for the corresponding channel becomes 49.9kΩ.
- Jumper **SB9** connects the digital **GND** and analog **AGND** planes at a single point. Keeping analog and digital grounds separate elsewhere on the board prevents digital switching noise from coupling into the sensitive analog measurement circuits; SB9 provides the one intentional tie-point between them. By default, the jumper is closed — this is the correct setting for standalone use, ensuring the analog ground has a return path. Open this jumper only if the module is used alongside other boards where AGND is already tied to GND elsewhere in the system (to avoid a ground loop).

## CONNECTING EXTERNAL SIGNALS

The **H1-H2** connectors on the module are used to connect external analog signals. The **H3-H4** connectors are for relay outputs. Connect via 3.5mm pitch removable terminal blocks. The pin layout is marked on the **bottom-side** of the module. When **using unipolar input mode**, polarity must be observed when connecting external input signals.

The following diagram shows the wiring configuration for both voltage and current measurement. This wiring configuration can be done for any channel, regardless of the commutation for other channels.




## CONFIGURATION TABLES

The **bottom-side** of the PCB indicates the precision variant of the **IoTextra Combo** module:

- **Lite**
- **STD** (Standard)




The **bottom-side** also provides information about the possible relay model installed in the **IoTextra Combo** module.




Possible relay models:

- HF33F/005-ZS3
- Omron G5Q-14
- Panasonic ALQ105
- TE PE014005




It also provides information indicating whether the **Qwiic©** connectors for **I²C** are installed

## ACCESSORIES:

The following accessories may be required for using the module:

- A set of four removable terminal blocks with a 3.5mm pitch for terminal blocks **H1-H4**.
- A set of two standoffs and four screws for mounting the module into the **IoTbase** series base module.
- Cable for the **Qwiic©** connector.